Status	Finished
Started	Tuesday, 20 October 2026, 1:48 PM
Completed	Tuesday, 20 October 2026, 2:00 PM
Duration	12 mins 9 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It divides the list into two halves, sorts each half, and then merges them
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **2**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Bubble Sort
c. Linear Search
d. Quick Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. Each part is sorted independently
- c. The parts are merged without sorting
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Linear Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Sorting data in a non-sequential manner
c. Using the divide-and-conquer approach
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Quick Sort
c. Heap Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sort()

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Finding a specific element in a list
- c. Arranging data in a specific order
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Bubble Sort
- c. Binary Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Merge Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Greedy method
c. Dynamic programming
d. Backtracking

The correct answer is:
Divide and conquer

Question **14**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Searching
c. Merging
d. Complexity

The correct answer is:
Complexity

Question **15**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass